

$$\textcircled{*} \left(\frac{\partial G}{\partial p} \right)_T = ? \textcircled{**} \left(\frac{\partial H}{\partial p} \right)_S$$

Is there any relation b/n above two equations.

properties: S, P, V, T

Energy: H, G, A, U

Gibbs free energy
Gibbs-Helmholtz free energy

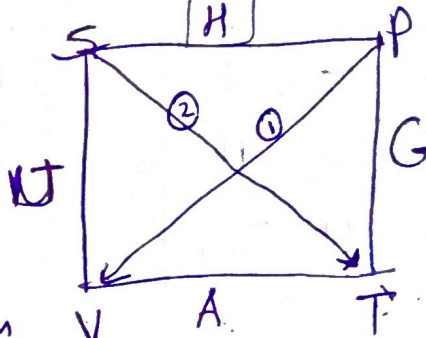
Thermodynamic relations:

$$dH = Tds + Vdp$$

$$dG = Vdp - SdT$$

$$dA = -pdv - SdT$$

$$dU = Tds - pdv$$



arrows are emerging from variables

from S
arrow is terminating at T, so -ve

emerging in = -ve
at the point
the other variable

Going to

$$Vdp - SdT$$

$$\textcircled{*} \left(\frac{\partial G}{\partial p} \right)_T = \left(\frac{\partial G}{\partial p} \right)_T$$

Whatever given, write the first one as it is

$$= \frac{Vdp}{dp} - S \left(\frac{dT}{dp} \right)_T$$

$$= V - S \left(\frac{dT}{dp} \right)_T$$

$$T = C$$

$$T_1 = T_2$$

$$T_2 - T_1 = 0$$

$$\Delta T = 0$$

$$dT = 0$$

$$\textcircled{**} \left(\frac{\partial H}{\partial p} \right)_S = Tds + Vdp$$

$$\Rightarrow \left(\frac{\partial H}{\partial p} \right)_S = V$$

$$\Rightarrow \left(\frac{\partial G}{\partial p} \right)_T = \left(\frac{\partial H}{\partial p} \right)_S$$

Maxwell Relations Trick

- Maxwell ^{H, G, A, U} proved that energy is state function by the help of Euler theory.

Good Professors Have studied Under Very Fine Teachers

Gibbs

Pressure

Enthalpy

Entropy

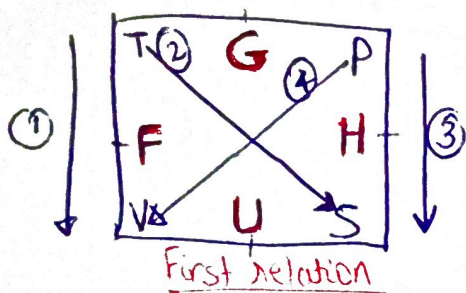
Internal Energy

Volume

Helm. Free Energy

Temp.

#1



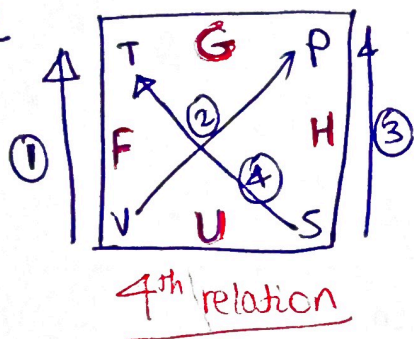
$$\left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial P}{\partial S}\right)_V$$

starting point of arrow
diagonal reach of arrow

⊛ Where to put -ve sign

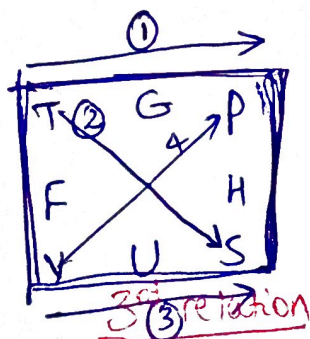
⊛ Whenever, both S & P come together, write negative sign on that side.

#2



$$\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$$

#3

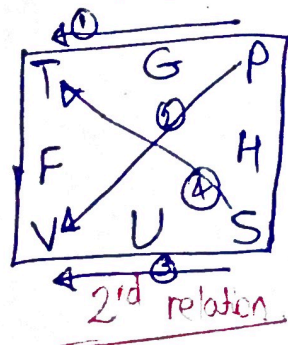


$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

⊛ Both have P & S together

#4



$$\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

⇒ No need of -ve signs, since neither of the terms has both S & P at same time

Cyclic relation for ideal gas equation:

$$\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = -1$$

For $pV = RT$

$\uparrow$ $\uparrow$ $\uparrow$
 y x z

$$\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1$$

$$\frac{R}{V} * \frac{1}{R} * \frac{-RT}{p^2} =$$

And $pV = RT$

$$\Rightarrow -1 = -1$$

$$\frac{\partial \left(\frac{1}{x}\right)}{\partial x}$$

$$\frac{\partial x^{-1}}{\partial x}$$

$$-1(x^{-1-1}) = -x^{-2} = -\frac{1}{x^2}$$

$$\left[\frac{d}{dx} x^n = nx^{n-1} \right]$$

$$p = \frac{RT}{V}$$

$$T = \frac{pV}{R}$$

$$V = \frac{RT}{p}$$

$$\left. \frac{dp}{dT} \right|_V = \frac{R}{V}$$

$$\left. \frac{\partial T}{\partial V} \right|_p = \frac{p}{R}$$

$$\left. \frac{\partial V}{\partial p} \right|_T = -\frac{RT}{p^2}$$

$$\left. \frac{\partial \left(\frac{RT}{p}\right)}{\partial p} \right|_T = \partial \left(\frac{R}{p}\right)$$

$$RT \left(\frac{\partial \left(\frac{1}{p}\right)}{\partial p} \right)_T = f(x, y)$$

$$g(x, y)$$

$$V = \left(\frac{RT}{p}\right)(T)$$

$$\frac{\partial z}{\partial x} = \frac{\frac{\partial f}{\partial x} g(x, y) - \frac{\partial g}{\partial x} f(x, y)}{(g(x, y))^2}$$

$$\left(\frac{\partial y}{\partial T}\right)$$

$$= \frac{\frac{\partial f}{\partial y} \cdot g(x, y) - \frac{\partial g}{\partial y} \cdot f(x, y)}{(g(x, y))^2}$$

$$\Rightarrow \frac{\left(\frac{\partial (RT)}{\partial T}\right)_p \cdot p - \left(\frac{\partial p}{\partial T}\right)_T \cdot RT}{p^2}$$

$$p^2$$

for a property $dx = 0$

Example

Reciprocity relation

$$pV = RT$$

$\uparrow \quad \uparrow \quad \uparrow$
 $z \quad x \quad y$

$$z = z(x, y)$$

$$\left. \frac{\partial x}{\partial y} \right|_z$$

$$p = \frac{RT}{V}$$

$$V = \frac{RT}{p}$$

$$T = \frac{pV}{R}$$

Partial derivative of x
with respect to y

holding z constant.

$$\left. \frac{\partial x}{\partial y} \right|_z = \left. \frac{\partial V}{\partial T} \right|_p = \frac{\partial \left(\frac{RT}{p} \right)}{\partial T} \bigg|_p = \frac{R}{p}$$

$$\left. \frac{\partial y}{\partial x} \right|_z = \left. \frac{\partial T}{\partial V} \right|_p = \frac{\partial \left(\frac{pV}{R} \right)}{\partial V} \bigg|_p = \frac{p}{R}$$

$$\Rightarrow \left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial x} \right)_z = 1$$

$$\frac{R}{p} \times \frac{p}{R} = 1$$

Check reciprocity relation,
using the ideal gas equation.

$$\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial x} \right)_z = 1$$

so for $pV = RT$

$\uparrow \quad \uparrow \quad \uparrow$
 $z \quad x \quad y$

$$\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial x} \right)_z = 1$$

$\uparrow \quad \uparrow$
 $T \quad V$

$$\Rightarrow \left. \frac{\partial y}{\partial T} \right|_p \times \left. \frac{\partial T}{\partial V} \right|_p =$$

$$\boxed{V = \frac{RT}{p}}$$

$$\left. \frac{\partial V}{\partial T} \right|_p = \frac{R}{p}$$

$$\boxed{T = \frac{pV}{R}}$$

$$\left. \frac{\partial T}{\partial V} \right|_p = \frac{p}{R}$$

$$\Rightarrow \frac{R}{p} \times \frac{p}{R}$$

$$= 1 //$$

Maxwell Relations

Maxwell proved that energy is state function by the help of Euler theorem.

Path function

↓
dependent on path

E.g. w, q

$$\oint w, q \neq 0$$

State function

↓
dependent on initial & final state

H, G, A, U

$$\oint U = 0$$

Since its cyclic loop, initial & final states are same.

$$dH = Tds + vdp \Rightarrow \left(\frac{dT}{dp}\right)_s = \left(\frac{dv}{ds}\right)_p$$

$$dG = vdp - sdT \Rightarrow \left(\frac{dv}{dT}\right)_p = -\left(\frac{ds}{dp}\right)_T$$

$$dA = -pdv - sdT \Rightarrow \left(\frac{dp}{dT}\right)_v = \left(\frac{ds}{dv}\right)_T$$

$$dU = Tds - pdv \Rightarrow \left(\frac{dT}{dv}\right)_s = \left(\frac{-dp}{ds}\right)_v$$

Euler Theory. Arbitrary variables

$$df = Adz + ydx$$

$$\left(\frac{dA}{dx}\right)_z = -\left(\frac{dy}{dz}\right)_x$$

Euler theorem

Example

How to check?

1. $PV = ST$ cross multiplication
2. Check operator & constant

Example $\left(\frac{dU}{dV}\right)_T = ?$ for ideal gas

$$dU = Tds - pdv$$

$$\left(\frac{dU}{dV}\right)_T = T\left(\frac{ds}{dV}\right)_T - \cancel{p\left(\frac{dV}{dV}\right)_T}$$

$$= T\left(\frac{ds}{dV}\right)_T - p$$

for ideal gas, $PV = RT \Rightarrow p = \frac{RT}{V}$

By applying maxwell eqn

$$\left(\frac{ds}{dV}\right)_T = \left(\frac{dp}{dT}\right)_V$$

$$\Rightarrow \left(\frac{dU}{dV}\right)_T = T\left(\frac{dp}{dT}\right)_V - p$$

$$= T\left(\frac{d\left(\frac{RT}{V}\right)}{dT}\right)_V - p$$

$$= \frac{TR}{V}\left(\frac{d(T)}{dT}\right) - p$$

$$= \frac{TR}{V} - \frac{RT}{V}$$

$$= \underline{\underline{0}}$$